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DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

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Total Marks:_________

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Project Proposal

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Library Management System

1. Project Introduction

This project presents a Library Management System developed in 8086 Assembly Language using the EMU 8086 emulator. The system demonstrates low-level programming concepts including memory management, interrupt handling, string operations, and direct hardware interfacing through DOS interrupts (INT 21h). The project is designed to meet university lab requirements for the Assembly Language course.

2. Project Objectives

- Implement a fully functional menu-driven library system in x86 16-bit Assembly
- Use INT 21h DOS interrupts for input/output and screen management
- Manage book records using in-memory data structures (arrays in the data segment)
- Demonstrate use of procedures, loops, conditional jumps, and string operations
- Provide Add, Search, Display All, Delete, and Issue/Return book functionalities
- Display user-friendly text-based interface using BIOS/DOS calls

3. Project Scope

3.1 In Scope

- Add new book (Title + Book ID + Author)
- Search book by ID
- Display list of all books
- Delete a book record

- Issue book to student and mark as unavailable
- Return book and mark as available
- Main menu navigation loop

3.2 Out of Scope

- File storage / persistent data (data is in-memory only)
- GUI or graphical interface
- Network or multi-user support

4. Tools & Technologies

Component	Tool / Technology
Emulator	EMU 8086 (x86 16-bit emulator)
Language	Intel x86 Assembly Language (16-bit)
OS Interface	DOS Interrupts (INT 21h, INT 10h)
Architecture	8086 Processor (1MB address space, 16-bit registers)
IDE / Editor	EMU 8086 built-in editor
Testing	EMU 8086 step-by-step debugger

5. System Design

5.1 Memory Layout (Data Segment)

The data segment stores all book records as fixed-size arrays. Each book occupies a fixed block: 30 bytes for title, 10 bytes for author, 5 bytes for book ID, and 1 byte for availability flag.

5.2 Registers Used

Register	Purpose
AX	Function codes for INT 21h, arithmetic
BX	Base address for memory indexing
CX	Loop counter, string length
DX	Offset address for string output (INT 21h/09h)
SI / DI	Source / Destination index for string ops
BP	Base pointer for stack frame

5.3 Key Procedures

- MAIN_MENU — displays menu and reads user choice
- ADD_BOOK — stores new book data in next free slot
- SEARCH_BOOK — linear search by Book ID using CMPSB
- DISPLAY_ALL — iterates through all records and prints
- DELETE_BOOK — zeroes out a record by ID
- ISSUE_BOOK — sets availability flag to 0
- RETURN_BOOK — resets availability flag to 1
- PRINT_STRING — reusable DX+INT 21h/09h output routine
- READ_STRING — reads keyboard input using INT 21h/0Ah
- CLEAR_SCREEN — uses BIOS INT 10h to clear display

6. Program Flow

START → Initialize Data Segment → Clear Screen → Show Main Menu → Read User Input

Option 1 → ADD_BOOK → Return to Menu

Option 2 → SEARCH_BOOK → Display Result → Return to Menu

Option 3 → DISPLAY_ALL → Return to Menu

Option 4 → DELETE_BOOK → Return to Menu

Option 5 → ISSUE_BOOK → Return to Menu

Option 6 → RETURN_BOOK → Return to Menu

Option 7 → Exit (INT 21h / 4Ch)

7. Project Schedule

Phase	Task	Duration
Phase 1	Requirements Analysis & Design	1 Day
Phase 2	Data Segment & Memory Layout	2 Hours
Phase 3	Menu System & I/O Routines	3 Hours
Phase 4	Add / Display / Search Features	3 Hours
Phase 5	Delete / Issue / Return Features	2 Hours
Phase 6	Testing & Debugging in EMU 8086	2 Hours
Phase 7	Final Review & Documentation	1 Hour

8. Expected Output

The program will display a text-based main menu. User can type a number (1-7) to select an operation. Results (book list, search result, status messages) will be displayed on screen using DOS print calls. All operations will work within EMU 8086 without requiring any external library or file system access.

9. Constraints & Limitations

- Maximum 10 books can be stored (expandable by increasing array size in data segment)
- Book titles limited to 30 characters, Author to 10 characters, ID to 5 characters
- Data is lost when the program exits (no file I/O)
- Runs only in Real Mode (16-bit) within EMU 8086 emulator

10. Conclusion

This Library Management System in EMU 8086 Assembly Language will effectively demonstrate core concepts of low-level programming: register-based computation, direct memory access, DOS interrupt calls, and structured procedure design. The project is feasible within the lab timeline and fulfills all requirements of the Assembly Language course project.